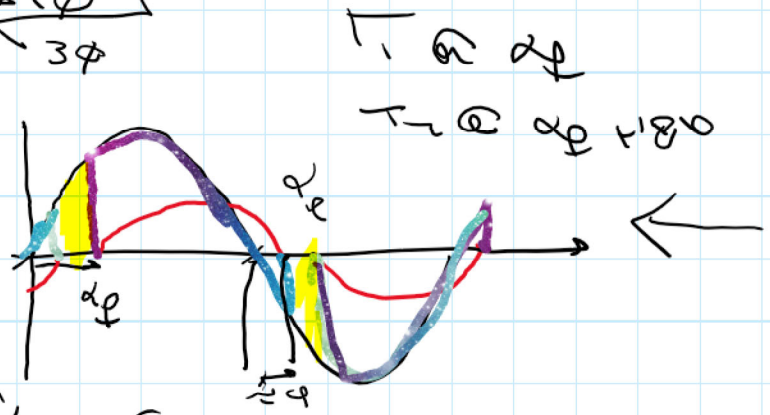
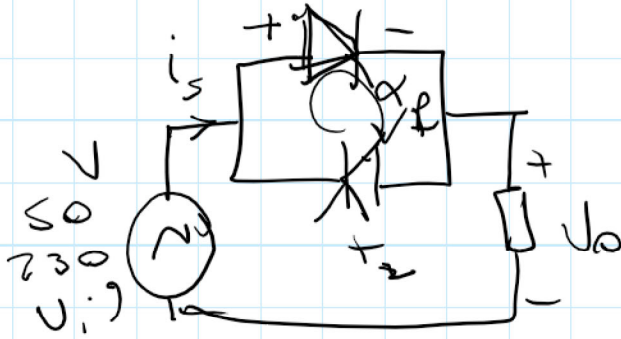


Recall

AC-AC Converters $\begin{cases} \text{AC Voltage controllers (} V_{rms} \text{)} \\ \text{Cycloconverter (} V_{rms} \text{)} \end{cases}$

AC Voltage controllers $\begin{cases} \text{1 } \phi \\ \text{3 } \phi \end{cases}$

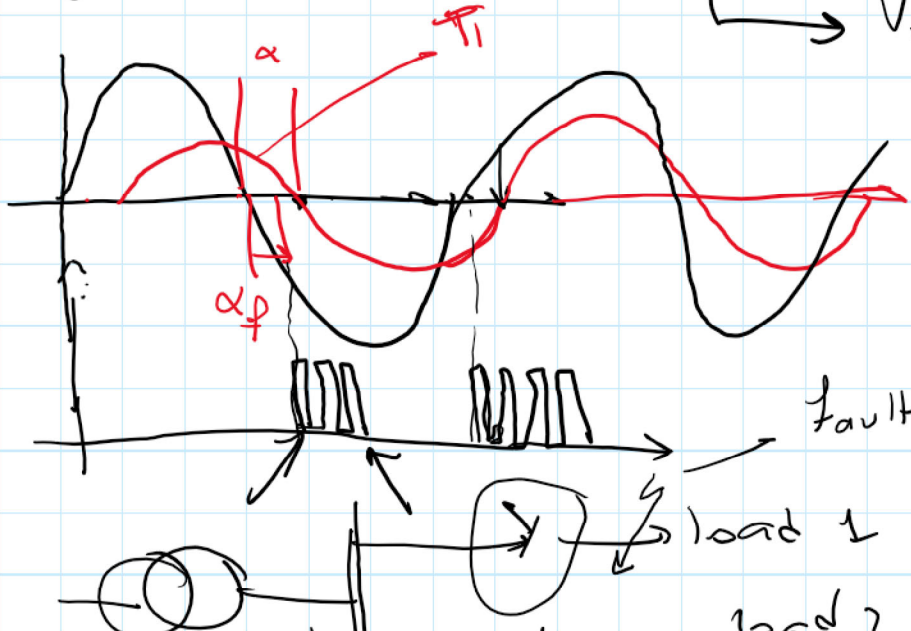
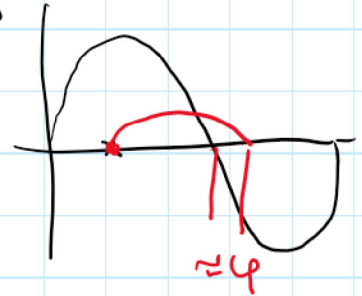


$$V_o = \left[\frac{1}{\pi} \int_{\alpha}^{\alpha+\pi} (V_m \sin \omega t)^2 d\omega t \right]^{1/2} = V_m \left[\frac{1}{\pi} \left\{ \delta - \alpha - \frac{1}{2} \sin 2\delta + \sin 2\alpha \right\} \right]^{1/2}$$

$$\frac{V_o}{V_i} = m = \left[\frac{1}{\pi} \left\{ \delta - \alpha - \frac{1}{2} (\sin 2\delta - \sin 2\alpha) \right\} \right]^{1/2}$$

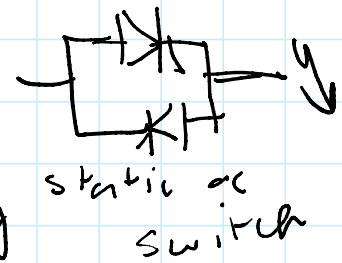
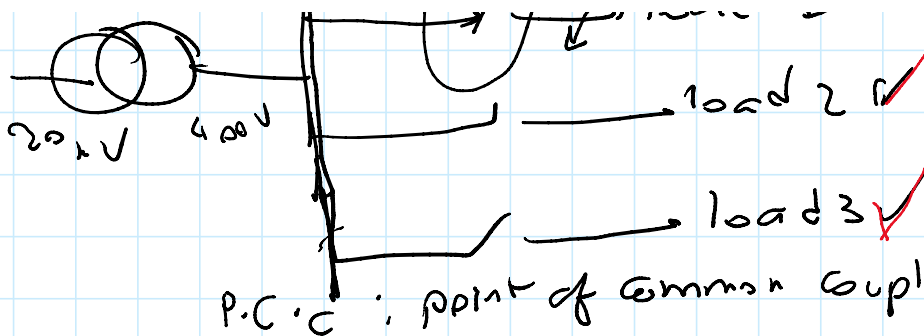
Observations:

- ① load current is always discontinuous.
- ② if $\alpha < \phi$ (load angle)



$V_{rms \text{ max}} = V_o$
ac switch
static





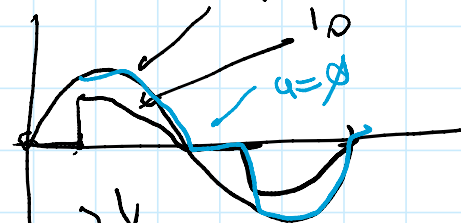
$$\bar{V}_o = \frac{3}{\pi} V_{mp} \cos \phi$$

$$M = \frac{V_o}{V_i} = \left[\frac{1}{\pi} \left(\alpha_e - \phi - \frac{1}{2} (\sin 2\alpha_e - \sin 2\phi) \right) \right]^{1/2}$$

$$(3) m = f(\alpha_e, \phi) \Rightarrow f(\alpha_e, \phi)$$

load is resistive, $\phi = 0$

$$\alpha_e = \pi$$

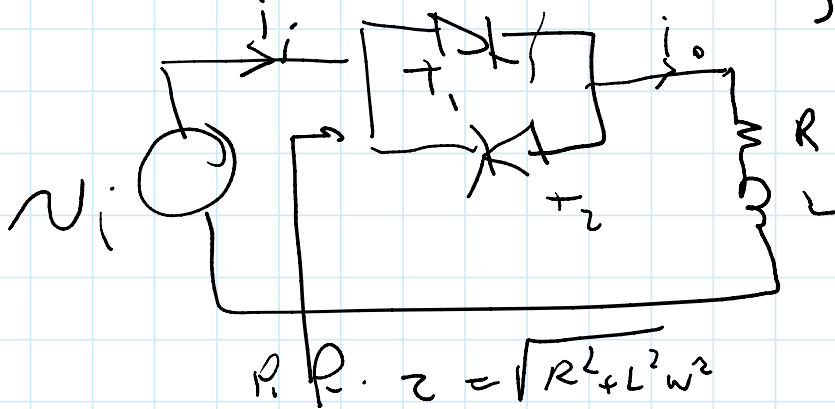
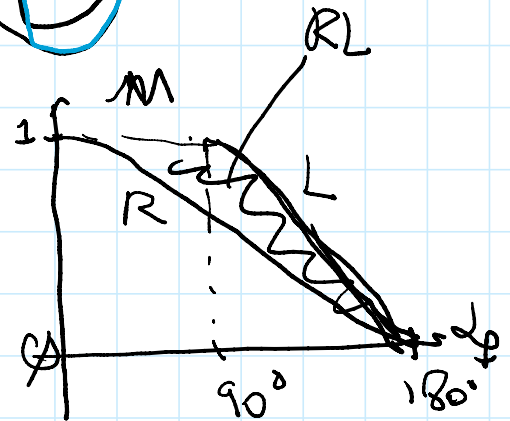


$$M = \left[\frac{1}{\pi} \left(\pi - \alpha_e + \frac{1}{2} \sin 2\alpha_e \right) \right]^{1/2}$$

load is pure inductive ($\phi = 90^\circ$)

$$\alpha_e = \pi - \phi$$

$$M = \left[\frac{1}{\pi} \left(\pi - \alpha_e + \frac{1}{2} \sin 2\alpha_e \right) \right]^{1/2}$$



$$P.C.C. Z = \sqrt{R^2 + L^2 \omega^2}$$

$$\begin{aligned} P.f. &= \frac{P_i}{S_i} = \frac{P_o}{S_i} = \frac{R I_o^2}{V_i I} \\ &= \frac{R I_o}{V_i} = \frac{Z \cos \phi I_o}{V_i} \\ &= \frac{I_o}{V_i / Z} \cos \phi \end{aligned}$$

$$P.f. = \frac{I_o}{I_{o \max}} \cos \phi \ll \cos \phi$$

normal :

$$I_{rms} = \frac{I_{max}}{\sqrt{2}}$$

example 1:

an ac-voltage controller is used in a theatre to control the light find the reduction in power when $\alpha_f = 60^\circ$.

resistive load: $m = \frac{V_o}{V_i} = \left[\frac{1}{\pi} (\pi - \alpha_f + \frac{1}{2} \sin 2\alpha_f) \right]^{\frac{1}{2}} = 0.897$
 $\alpha_f = 60^\circ, m = 1/3$

$$P \propto V_o^2 \propto 0.897^2 = 0.805$$

19.5% reduction in power

example 2:

ac-voltage controller, RL load, $R = 2.5 \Omega$, $L = 6.5 \text{ mH}$
 $V_i = 120 \text{ V}$, $f_s = 60 \text{ Hz}$, $\alpha_f = \pi/2$

a) Conduction angle? $\beta = \alpha_c - \alpha_f$

$\alpha_c > \alpha_f \Rightarrow \text{discont.}$

$$\phi = \tan^{-1} \frac{\omega L}{R} = 44.413^\circ < 90^\circ \text{ w.t. } 90^\circ$$

$$i_o(\omega t) = \frac{120\sqrt{2}}{3.5} \left\{ \sin(\omega t - 44.413^\circ) - e^{-\frac{R}{\omega L} \sin(\omega t - 44.413^\circ)} \sin(44.413^\circ) \right\}$$

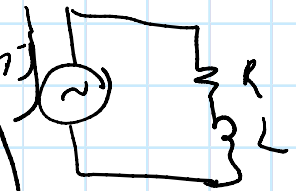
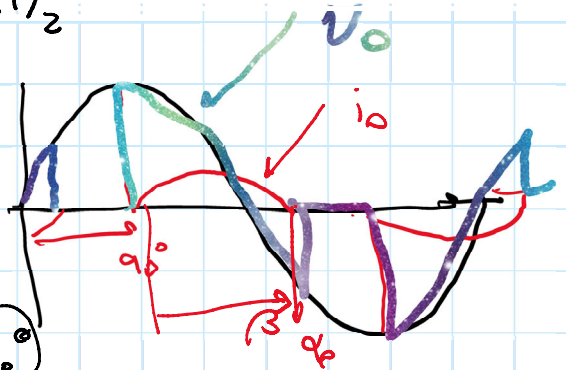
$$Z = \sqrt{R^2 + \omega^2 L^2} = 3.5$$

$$R i_o + L \frac{di_o}{dt} = V_m \sin \omega t$$

$$i_o(\omega t) = 0 \rightarrow \omega t = \pi/2 = \alpha_f$$

$$\omega t = 220.35$$

$$\beta = 220.35 - 90 = 130.65^\circ$$



b) $V_o = ?$

$$m = \left[\frac{1}{\pi} (\pi - \alpha_f - \frac{1}{2} \sin 2\alpha_f) \right]^{\frac{1}{2}}$$

$$V_o = m V_i = 68.09 \text{ V}$$

c) $\bar{I}_T = ? = \frac{1}{\pi} \int_{\alpha_f}^{\alpha_c} i_o(\omega t) d\omega t = 8.25 \text{ A}$

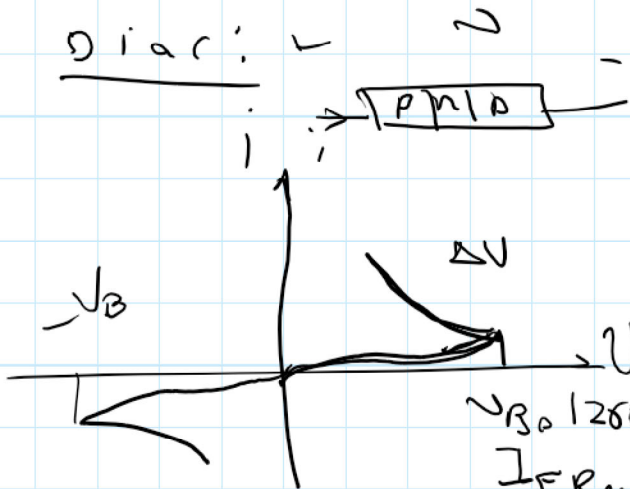
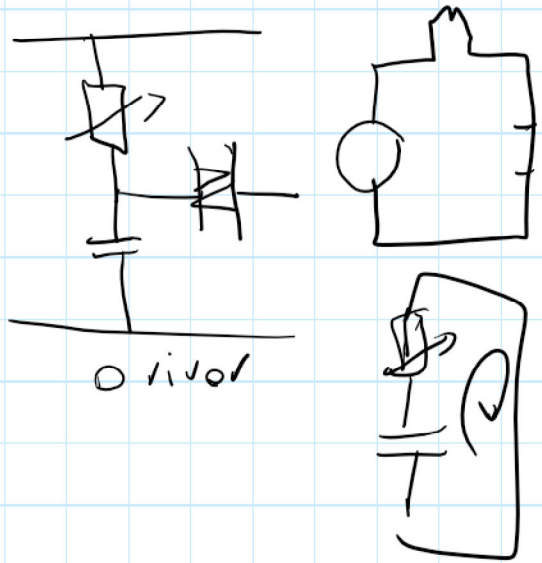
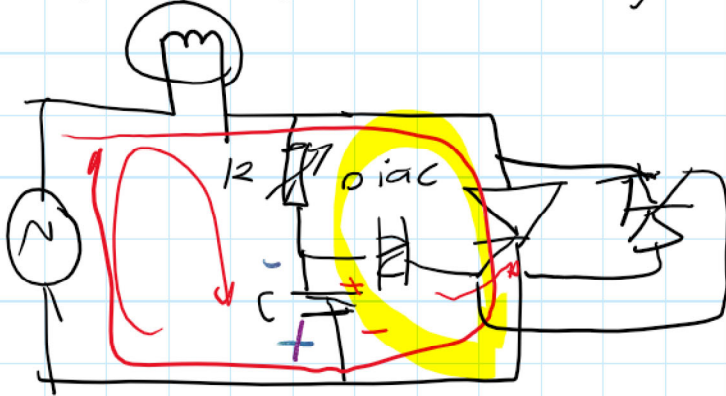
$$c) \overline{I_T} = \frac{1}{2\pi} \int_{\omega_f}^{2\omega_f} i_0(\omega T) d\omega T = 8.25 A$$

$$a) I_0 = ? = \sqrt{2} I_T = \sqrt{2} \times 15.07 = 21.3 \text{ A}$$

$$I_T = \left[\frac{1}{2\pi} \int_{\phi_f}^{\phi_g} \frac{2}{10(\omega t)} d\omega t \right]^{1/2} = 15.07 \mu$$

$$e) P_i = ? = \frac{R_{010}}{V_i} = \frac{2.5 \times 21.3}{120} = 0.444 \text{ (log)}$$

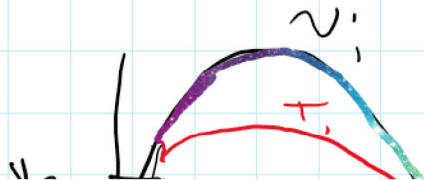
example ③: light density control



$V_{BO}(28V-36V) \rightarrow$ Break-over Voltage

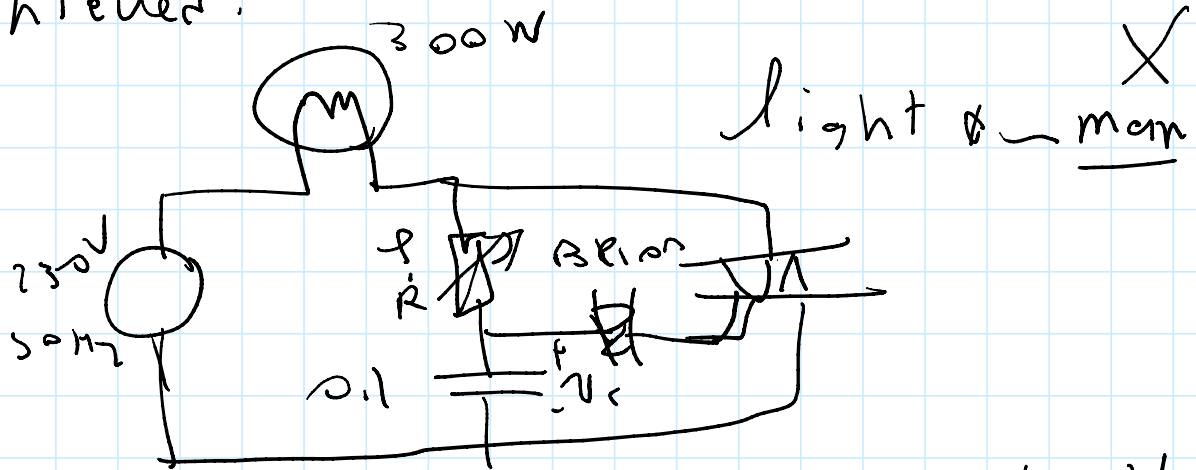
I_{FRM} : maximum repetitive forward current

DV : break-back voltage $> 6V$


$$\epsilon_{nc}/\epsilon_{ni}$$

- determine R such that a full-light density control is achieved.

$$I_{\text{arm}} = 2A$$



R must be such that V_c never reaches $V_{B0 \min}$

$$|V_c| = \left| \frac{1/j\omega C}{R + 1/j\omega C} \right| V_m \sin \omega t \Rightarrow V_{c \max} = \frac{V_m}{\sqrt{1 + R^2 C^2 \omega^2}} \quad R \gg R_{\text{lamp}}$$

$$V_{c \max} < V_{B0 \min}$$

if $R^2 C^2 \omega^2 \gg 1 \Rightarrow V_{c \max} \approx \frac{V_m}{RC\omega} > V_{B0 \min}$

$$R > \frac{V_m}{C\omega V_{B0 \min}} = \frac{230\sqrt{2}}{0.1 \times 10^{-6} \times 2\pi \times 50 \times 28} = 354 \Omega$$

$R = 400 \Omega$